MAT1100 Homework 3

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Question 4

(a) *Proof.* Suppose $a + b\sqrt{-5}, c + d\sqrt{-5} \in R$ then

$$N\Big((a + b\sqrt{-5})(c + d\sqrt{-5})\Big) = N\Big((ac - 5bd) + (ad + bc)\sqrt{-5}\Big) = (ac - 5bd)^2 + 5(ad + bc)^2$$

and

$$\begin{aligned} N(a + b\sqrt{-5})N(c + d\sqrt{-5}) &= (a^2 + 5b^2)(c^2 + 5d^2) \\ &= (ac)^2 + 5(bc)^2 + 5(ad)^2 + (5bd)^2 \\ &= (ac)^2 + 5(bc)^2 + 5(ad)^2 + (5bd)^2 + 10abcd - 10abcd \\ &= (ac - 5bd)^2 + 5(ad + bc)^2 \end{aligned}$$

hence N is multiplicative.

Now I claim that $\alpha = a + b\sqrt{-5} \in R$ is a unit if and only if $N(\alpha) = 1$. Suppose that α is a unit then

$$1 = N(1) = N(\alpha^{-1}\alpha) = N(\alpha^{-1})N(\alpha)$$

and since $N \in \mathbb{Z}^{\geq 0}$ this implies that $N(\alpha) = 1$. Conversely, suppose $N(\alpha) = 1$, observe that $N(\alpha) = \alpha\overline{\alpha}$ this implies that $\overline{\alpha} = a - b\sqrt{-5} \in R$ is the multiplicative inverse of α.

Thus to compute the units of R it suffices to find all of the elements of unit norm. In other words, all $a, b \in \mathbb{Z}$ such that $a^2 + 5b^2 = 1$ and clearly the only possible pairs are $(\pm 1, 0)$ and therefore the only units in R are ± 1. □

(b) *Proof.* First a useful fact is that in R there are no elements of norm 2 or 3. To see this it suffices to show that the equations

$$a^2 + 5b^2 = 2 \quad a^2 + 5b^2 = 3$$

have no solutions over $\mathbb{Z}$. This is clear as for $b > 0$ we have $5b^2 > 3, 2$ forcing $b = 0$ but then $a^2 = 2$ and $a^2 = 3$ have no solutions over the integers. Now $N(2) = 4$ and if $\alpha, \beta \in R$ such that $2 = \alpha\beta$ then

$$N(\alpha)N(\beta) = 4$$

and since there are no elements of norm 2 this implies that either $N(\alpha) = 1$ or $N(\beta) = 1$ i.e. either α or β is a unit and hence 2 is irreducible.

Similarly $N(3) = 9$, $N(1 \pm \sqrt{-5}) = 6$ and a similar argument shows that these elements are irreducible. Thus 6 factors in two ways as a product of irreducibles

$$6 = 2 \cdot 3 = (1 + \sqrt{-5})(1 - \sqrt{-5})$$

Now since the only units in R are ± 1 these factorizations are distinct as an element with non-trivial imaginary part cannot be associates with a real valued element. Thus R is not a UFD as 6 does not have a unique factorization into a product of irreducibles. □

(c) *Proof.* Let $\alpha = \sqrt{-5}$ for convenience. Now to show that I is a proper ideal, it suffices to show that $1 \notin I$. To see this, note that if $1 \in I$ then there exist $s, t \in R$ such that

$$1 = 2s + (1 + \alpha)t$$

and taking norms this implies

$$1 = 4N(s) + 6N(t)$$

and since the norm is non-negative this is not possible. Thus the quotient ring R/I is not the zero ring.

Now to see that I is maximal it suffices to show that R/I is a field. Suppose $a + b\alpha \in R$ let $\alpha = \lfloor a/2 \rfloor$ and $\beta = \lfloor b/2 \rfloor$ then $a - 2\alpha, b - 2\beta \in \{0, 1\}$ and

$$a + b\alpha = a + b\alpha - 2(\alpha + \beta\alpha) \mod I$$

and hence R/I has at most 4 elements possibly represented by $0, 1, \alpha, 1 + \alpha$. Now

$$0 = 1 + \alpha \mod I$$

and

$$1 = 1 - (1 + \alpha) = -\alpha = \alpha \mod I$$

thus R/I has at most 2 elements and since R/I is not the zero ring and R is unital this implies that $R/I \cong \mathbb{F}_2$. Hence I is maximal.

Finally to see that I is not principal suppose for contradiction that there exists $c \in R$ such that $(c) = I$. Then $c \mid 2$ and $c \mid 1 + \alpha$ which implies that that $N(c) \mid 4$ and $N(c) \mid 6$ hence $N(c) \in \{1, 2\}$. Now since no elements in R have norm 2 this implies that $N(c) = 1$ and it is a unit. Thus

$$R = (c) = I$$

but this is a contradiction as it was shown that I was maximal and hence proper. □